

**Time:** 5 minutes**Outcome:** Calculate multiplier effects and evaluate crowding out.**Difficulty:** ●●● Advanced

## Fiscal Multipliers and Crowding Out



### THE CORE IDEA

A fiscal expansion can generate further income-consumption rounds. The marginal propensity to consume (MPC) is the share of extra disposable income spent. With fixed prices, lump-sum taxes, and no import leakage, the spending multiplier is  $1 / (1 - \text{MPC})$  and the tax multiplier is  $-\text{MPC} / (1 - \text{MPC})$ . Higher interest rates can offset part of the expansion by crowding out private investment.



### HOW TO RECOGNIZE IT

- Separate the initial spending change from induced consumption rounds.
- A tax cut affects consumption only through the share of extra income spent.
- Higher interest rates reduce private investment, offsetting part of the gross expansion.

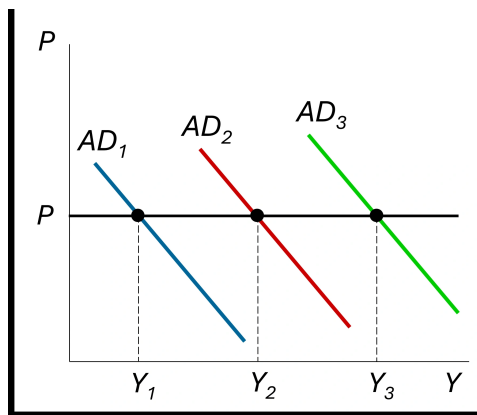


### WATCH OUT

A common mistake is to report a gross multiplier effect without subtracting private spending displaced by higher interest rates. Crowding out makes the net demand increase smaller.



### WORKED EXAMPLE: MULTIPLIER VERSUS CROWDING OUT



With  $\text{MPC} = 0.8$ , the simple spending multiplier is 5. A \$20 billion increase in purchases raises demand by \$100 billion before offsets. In the graph,  $AD_1$  to  $AD_3$  is the gross expansion; crowding out leaves  $AD_2$ . At fixed  $P$ , real output demanded rises to  $Y_2$  rather than  $Y_3$ . Actual equilibrium output also depends on aggregate supply.



### CHECK YOURSELF

Why might a tax cut have a smaller immediate effect than an equal increase in government purchases?

**READY?**

Return to the game and master this concept.